QQ-A-250/29A
November 18, 1977
SUPERSEDING
Int. Fed. Spec.
QQ-A-00250/29 (Navy-AS)
March 29, 1972

FEDERAL SPECIFICATION

ALUMINUM ALLOY 2124, PLATE

This specification was approved by the Commissioner, Federal Supply Services Administration, for the use of all Federal agencies.

(This specification sheet forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 2124 plate for intended use in the - T851 temper; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate shall be furnished in the - T351 or - T851 tempers, as specified, (see 6.2). The definition of these tempers shall be as specified in American National Standard ANSI H35.1.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following specifications and standards, of issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATION

QQ-A-250 -Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

FSC 9535

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FEDERAL STANDARDS

Fed. Std. No. 184 -Identification Marking of Aluminum, Magnesium and Titanium

Fed. Test Method Std. No. 151 - Metals; Test Methods

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

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MILITARY SPECIFICATIONS:

MIL-H-6088 -Heat Treatment of Aluminum Alloys

MIL-I-8950 -Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) STANDARD

ANSI H35.1 -Alloy and Temper Designation Systems for Aluminum

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

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3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition. 1/

Element	Percent	
	Minimum	Maximum
Copper	3.8	4.9
Magnesium	1.2	1.8
Manganese	0.30	0.9
Iron	-	0.30
Zinc	-	0.25
Silicon	-	0.20
Titanium	-	0.15
Chromium	-	0.10
Zirconium plus Titanium	-	0.20
Others, each	-	0.05
Others, total	-	0.15
Aluminum	Remainder	

1/ Analysis shall be made regularly only for the elements specifically mentioned in this table I. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Heat treatment. Heat treatment shall be in accordance with requirements for alloy 2024 plate in MIL-H-6088.

3.3 Mechanical properties.

3.3.1 Mechanical properties of -T851 material. The mechanical properties in the longitudinal, long transverse, and short transverse direction shall conform to the requirements of table II.

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TABLE II. Mechanical properties.

Thickness Inches	Direction of test	Tensile strength, minimum P.s.i.	Yield strength at 0.2 percent offset, minimum P.s.i.	Elongation 1/ in 2 in. or 4 times D, 2/ minimum Percent
1.000 thru 2.000	Longitudinal Long transverse Short transverse 3/	66,000 66,000 64,000	57,000 57,000 55,000	6 5 1.5
2.001 thru 3.000	Longitudinal Long transverse Short transverse	65,000 65,000 63,000	57,000 57,000 55,000	6 4 1.5
3.001 thru 4.000	Longitudinal Long transverse Short transverse	65,000 65,000 62,000	56,000 56,000 54,000	5 4 1.5
4.001 thru 5.000	Longitudinal Long transverse Short transverse	64,000 64,000 61,000	55,000 55,000 53,000	5 4 1.5
5.001 thru 6.000	Longitudinal Long transverse Short transverse	63,000 63,000 58,000	54,000 54,000 51,000	5 4 1.5

1/ Not required for material 1/2 inch or less in width
2/ D represents specimen diameter
3/ Short transverse specimens not required from material less than 1-1/2 inches in thickness.

3.3.2 Mechnical properties of -T351 material. Material supplied in the -T351 temper shall, after precipitation treatment, conform to the requirements of 3.3.1.

3.4 Resistance to stress corrosion cracking. When specified (see 6.2) plate in -T851 temper or -T351 after precipitation treatment shall be resistant to stress corrosion cracking when tested in accordance with 4.1.

3.5 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table III.

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TABLE III. Ultrasonic discontinuity acceptance limits. 1/

Thickness Inches	Maximum weight per piece 2/ Pounds	Discontinuity class 3/
1.000 thru 3.000	2,000	A
3.001 thru 4.500	2,000	B
4.501 thru 6.000	4/	4/

1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas.

2/ Class for weights in excess of those shown shall be as agreed upon by user and supplier.

3/ The discontinuity class limits are defined in MIL-I-8950.

4/ Shall be as agreed upon by user and supplier.

3.6 Marking. In addition to the marking required in Fed. Std. No. 184, plate shall be identified by a lot number marked in at least one location on each piece.

3. QUALITY ASSURANCE PROVISIONS (see QQ-A-250)

4.1 Resistance to stress-corrosion cracking. Specimens from each plate or plates produced from a single ingot shall be subjected to the following test:

(a) Thirty days exposure, using three 0.225 inch diameter tensile type specimens. Three C-ring specimens may be used for plate under 1.500 inches in thickness.

(b) Stressed in the short transverse direction with respect to grain flow and held at a constant strain. The stress level shall be 50 percent of the minimum long transverse yield strength specified in table II.

(c) The stressed specimens shall be exposed to a solution of 3-1/2 percent NaCl conforming to the purity and pH requirements of method 823 of Fed. Test Method No. 151. The pH, temperature, humidity controls and exposure cycle shall also conform to method 823.

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(d) Retesting for original specimen failure(s) at a rate of two specimens for each failed original is allowed. However all retest specimens must pass.

4.2 Mechanical property testing. Tensile tests shall be performed on specimens in the -T851 condition in accordance with QQ-A-250 except that tension test specimens shall be taken from all three grain directions as shown in 3.3.1.

5. 5. PREPARATION FOR DELIVERY (see QQ-A-250)

6. NOTES

6.1 Intended use. This material is intended for use in structural application requiring a minimum of residual stresses and improvement in short transverse properties over those exhibited by alloy 2024. Material in the -T351 temper provides greater formability and is intended for use after forming and precipitation heat treatment to the -T851 temper.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and including the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Dimensions required.

(c) Temper required (see 1.2.1).

(d) Requirements for sizes not specifically covered (see QQ-A-250).

(e) Requirement for testing stress corrosion cracking resistance (see 3.4).

(f) Whether ultrasonic inspection of plate is required (see 3.5).

(g) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Fracture toughness. Material producers and users are encouraged to accumulate fracture toughness (K_{IC}) data on this material as it is anticipated that future revision of this specification may contain fracture toughness requirements.

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DOCUMENT IDENTIFIER AND TITLE

QQ-A-250/29A ALUMINUM ALLOY 2124, PLATE

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